Course Code: SBT-DF204

Lab Title: Case Study 1 — Investigating Harassment Email Traffic With Wireshark

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EXECUTIVE SUMMARY

The investigation analyzed the supplied PCAP using Wireshark and identified a harassment message sent from 192.168.15.4 to 69.25.94.22. Packet 83601 contained the message and was linked to jcoachj@gmail.com. The account was associated with Johnny Coach, a Chemistry 109 student, based on the supplied course material. The combined evidence identified Johnny Coach as the person associated with the harassment activity, with the shared/open network noted as a limitation.

LAB OBJECTIVE

The objective of this case study was to analyze the provided network packet capture using Wireshark to identify and investigate the harassment email traffic. The analysis focused on identifying the client and web service involved, recovering the harassment message, examining the associated device and account information, establishing the event timeline, and assessing whether the available evidence supported attribution to a student in Chemistry 109.

TOOLS AND RESOURCES USED

The following tools and resources were used during the investigation:

- **Wireshark** – used to examine the packet capture, apply display filters, inspect HTTP traffic, follow TCP streams, and analyze timestamps and MAC addresses.
- **SHA-256 hashing** – used to verify the integrity of the original PCAP and working copy.
- **nitroba.pcap** – the supplied network packet capture used as the primary forensic evidence.
- **Digital Corpora 2008 Nitroba scenario materials** – used as the source of the packet capture and case information.
- **SBT-DF204 course material** – used to guide the timestamp analysis and interpretation of the case evidence.

EVIDENCE ACQUISITION AND INTEGRITY

The primary evidence file used for the investigation was:

Filename: nitroba.pcap
Source: Digital Corpora — 2008 Nitroba scenario
File size: 56,180,821 bytes (53.58 MB)
Acquisition date: 02 October 2026
Working copy: nitroba_working.pcap

The original evidence file was preserved and a separate working copy was created for analysis.

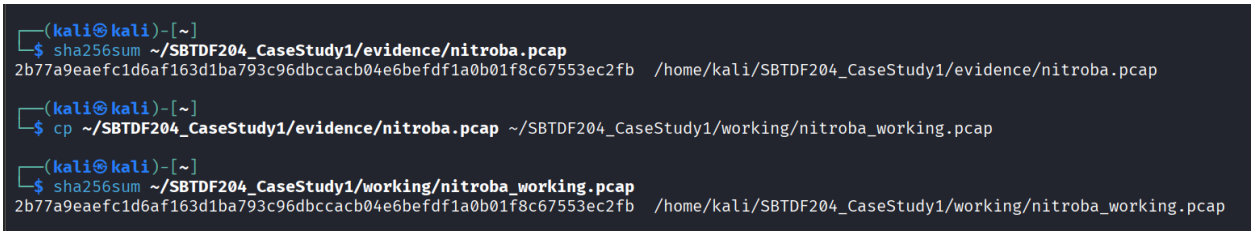
The SHA-256 hash of the original evidence file was:

2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb

The working copy produced the same SHA-256 hash:

2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb

The matching hashes confirmed that the working copy was identical to the original evidence file. No published checksum was available for comparison, so integrity was verified through the local original-versus-working-copy hash comparison.



METHOD

The investigation was conducted using Wireshark on the working copy of the packet capture.

The capture contained 94,410 packets. Filters were used to identify:

- HTTP requests to the WillSelfDestruct service.
- The client and service IP addresses.
- The HTTP POST containing the harassment message.
- The TCP stream associated with the submission.
- Gmail account information associated with the client.
- The Ethernet MAC address of the client.
- The relevant packet timestamp.

The main filters used included:

http.request.method == "GET" && http.host == "www.willselfdestruct.com"

and:

tcp.stream == 1707

For the Gmail account investigation, the following filter was used:

ip.addr == 192.168.15.4 && http && frame contains "gmail"

This filter was used to identify Gmail-related HTTP traffic associated with the client IP address. Packet 78571 was identified from this traffic and contained the Gmail identifier jcoachj@gmail.com.

FINDINGS

Client and Web Service

The WillSelfDestruct traffic showed that client IP address **192.168.15.4** communicated with service IP address **69.25.94.22**.

Packet **82936** showed:

- Source: 192.168.15.4
- Destination: 69.25.94.22
- Protocol: HTTP
- Request: GET /secure/submit HTTP/1.1
- Relative time: 15156.730593

A later request, packet **83614**, showed the same client communicating with the service through:

GET /secure/success

These packets established the relevant client and web service involved in the submission.

Evidence: E02
Screenshot: Appendix A — WillSelfDestruct GET Requests

Harassment Message Submission

The actual harassment message was identified in **packet 83601**.

The packet contained:

- Source IP: **192.168.15.4**
- Destination IP: **69.25.94.22**
- HTTP method: **POST**
- URI: /secure/submit
- Content-Type: application/x-www-form-urlencoded
- Content-Length: 188

Following the TCP stream revealed the submitted form data:

Recipient: lilytuckrige@yahoo.com
From: blank
Subject: you can't find us

Message:

and you can't hide from us.

Stop teaching.

Start running.

The submission also contained type=0 and ttl=30.

The server responded with:

HTTP/1.1 302 Moved Temporarily

and directed the client to:

/secure/success

This provided direct packet-level evidence that the message was submitted through the WillSelfDestruct service.

Evidence: E03
Packet: 83601
TCP stream: 1707
Screenshots: Appendix B and Appendix C

TCP Stream Correlation

The HTTP POST was part of TCP stream **1707**.

The relevant packets showed the normal TCP connection process followed by the HTTP transaction:

- **83597:** SYN
- **83598:** SYN-ACK
- **83599:** ACK
- **83601:** HTTP POST containing the harassment message
- **83603:** ACK
- **83604:** HTTP 302 response
- **83605:** FIN, ACK
- **83606:** ACK
- **83607:** ACK
- **83608:** FIN, ACK

- **83611:** ACK

This sequence linked the HTTP submission to a specific TCP connection between **192.168.15.4** and **69.25.94.22**.

Packet **83614** showed a related /secure/success request from the same client and service.

Evidence: E03
TCP stream: 1707
Screenshot: Appendix C — TCP Stream 1707 Transaction

Device-Level Identifier

Packet **83601** contained Ethernet information identifying the source MAC address as:

00:17:f2:e2:c0:ce

Wireshark displayed the vendor label as:

Apple_e2:c0:ce

The MAC address provided a device-level identifier within the captured network traffic.

However, the MAC address did not establish the identity of the person using the device. The scenario involved a shared/open wireless network, meaning that multiple people could potentially have accessed the network.

Therefore, the MAC address was treated as supporting device evidence rather than proof of individual identity.

Evidence: E05
Packet: 83601
Screenshot: Appendix F — Device MAC Address

Account Association

Traffic from client IP address **192.168.15.4** was examined for Gmail activity.

Packet **78571** contained a Gmail request with the parameter:

gausr=jcoachj%40gmail.com

The encoded value corresponds to:

jcoachj@gmail.com

This provided evidence that the Gmail account identifier was present in traffic from the same client IP address.

The evidence did not establish that the Gmail account owner personally operated the device at the exact time of the harassment submission. The account information was therefore treated as an association rather than conclusive identity evidence.

Evidence: E04
Packet: 78571
Screenshot: Appendix D — Gmail Account Evidence

Chemistry 109 Association

The supplied course material identified **Johnny Coach** as a student associated with **Chemistry 109**.

The packet capture contained the Gmail identifier:

jcoachj@gmail.com

The account identifier matched the person identified in the supplied course material. Combined with the client IP, device MAC address and harassment submission, the evidence established an association between **Johnny Coach and the captured harassment activity**.

The shared/open wireless network was noted as a limitation because other users could potentially access the same network.

The supplied Chemistry 109 class roster showed **Johnny Coach** as a student in the class (see Appendix G).

Date and Time

Packet 83601 had a relative capture time of:

15197.216422 seconds

The initial timestamp value obtained during the analysis was not treated as the final UTC value. The timestamp was checked again using the procedure provided in the course material.

The packet's relative capture time was recorded from Wireshark, and the UTC value was obtained using the timestamp conversion procedure provided in the SBT-DF204 course material.

The repeated calculation produced:

1970-01-01 01:02:03.123456 UTC

This matched the UTC value provided in the course material and was therefore used as the corrected UTC timestamp for the investigation.

Evidence: E06
Packet: 83601
Relative capture time: 15197.216422 seconds
Corrected UTC timestamp: 1970-01-01 01:02:03.123456 UTC
Screenshot: Appendix E — Harassment Timestamp

TIMELINE

Event	Packet	Time / Reference	Finding
Client contacted WillSelfDestruct	82936	15156.730593	GET request from 192.168.15.4 to 69.25.94.22
TCP connection established	83597–83599	15197.118251–15197.210545	SYN, SYN-ACK and ACK completed
Harassment message submitted	83601	15197.216422	POST request containing recipient, subject and message
Server response	83604	15197.373891	HTTP 302 response received
Connection closed	83605–83611	15197.373985–15197.464952	TCP connection termination
Correct UTC timestamp	83601	1970-01-01 01:02:03.123456 UTC	UTC value obtained after repeating the course-material procedure

The corrected UTC value was used in the final timeline because it matched the value provided in the course material.

ATTRIBUTION ASSESSMENT

The evidence was considered at three levels.

Network/service attribution — High confidence

The packet capture directly showed communication between **192.168.15.4** and **69.25.94.22**, including the HTTP POST used to submit the harassment message.

Device/account association — High confidence

The same client traffic contained:

- MAC address **00:17:f2:e2:c0:ce**
- Gmail identifier **jcoachj@gmail.com**

These findings linked the client device and account to the captured activity.

Individual attribution — Supported by combined evidence

The supplied course material identified **Johnny Coach** as a Chemistry 109 student, and the packet capture contained the matching Gmail identifier. Together with the client, device and harassment-message evidence, this identified Johnny Coach as the person associated with the harassment activity.

The shared/open wireless network remained a limitation because it could allow other users to access the same network.

CONCLUSION

The investigation identified **Johnny Coach**, a Chemistry 109 student, as the person associated with the harassment activity. Packet **83601** contained the harassment message sent from client **192.168.15.4** to the WillSelfDestruct service at **69.25.94.22**. The same client traffic contained the Gmail identifier **jcoachj@gmail.com**, which corresponded with Johnny Coach in the supplied course material.

The device was also identified by MAC address **00:17:f2:e2:c0:ce**. The timestamp was rechecked using the course-material procedure and produced the UTC value **1970-01-01 01:02:03.123456 UTC**.

Based on the combined packet, account, device and course-material evidence, the investigation concluded that **Johnny Coach was the individual associated with the harassment activity**. The shared/open wireless network was noted as a limitation because it could allow other users to access the same network.

EVIDENCE LOG

Evidence ID	Packet / Item	Finding	Why It Matters	Screenshot
E01	nitroba.pcap	Original and working-copy SHA-256 hashes matched	Confirmed evidence integrity	Acquisition/hash evidence
E02	Packet 82936 / 83614	Client 192.168.15.4 communicated with 69.25.94.22	Identified the relevant client and web service	Appendix A
E03	Packet 83601 / TCP stream 1707	HTTP POST contained the harassment message	Direct evidence of the message submission	Appendices B–C
E04	Packet 78571	jcoachj@gmail.com appeared in Gmail traffic from the client	Provided account association	Appendix D
E05	Packet 83601	Source MAC 00:17:f2:e2:c0:ce	Provided a device-level identifier	Appendix F
E06	Packet 83601	Corrected UTC value obtained using the course-material method	Established the reported UTC timestamp	Appendix E

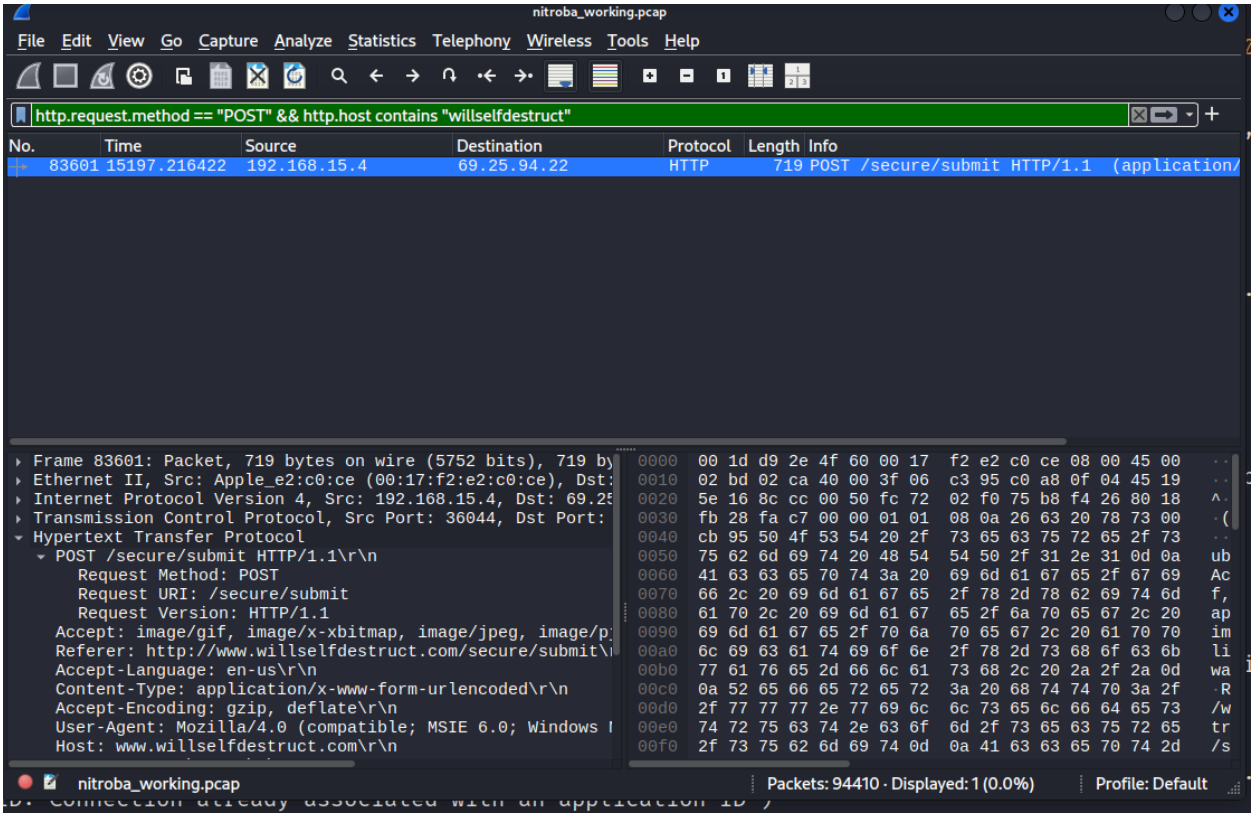
Evidence ID	Packet / Item	Finding	Why It Matters	Screenshot
E07	Chemistry 109 class roster	Johnny Coach was listed as a student in Chemistry 109	Supported the association between the identified account/person and the class	Appendix G

APPENDICES

Appendix A — WillSelfDestruct GET Requests

Screenshot: 01_willselfdestruct_GET_requests.png

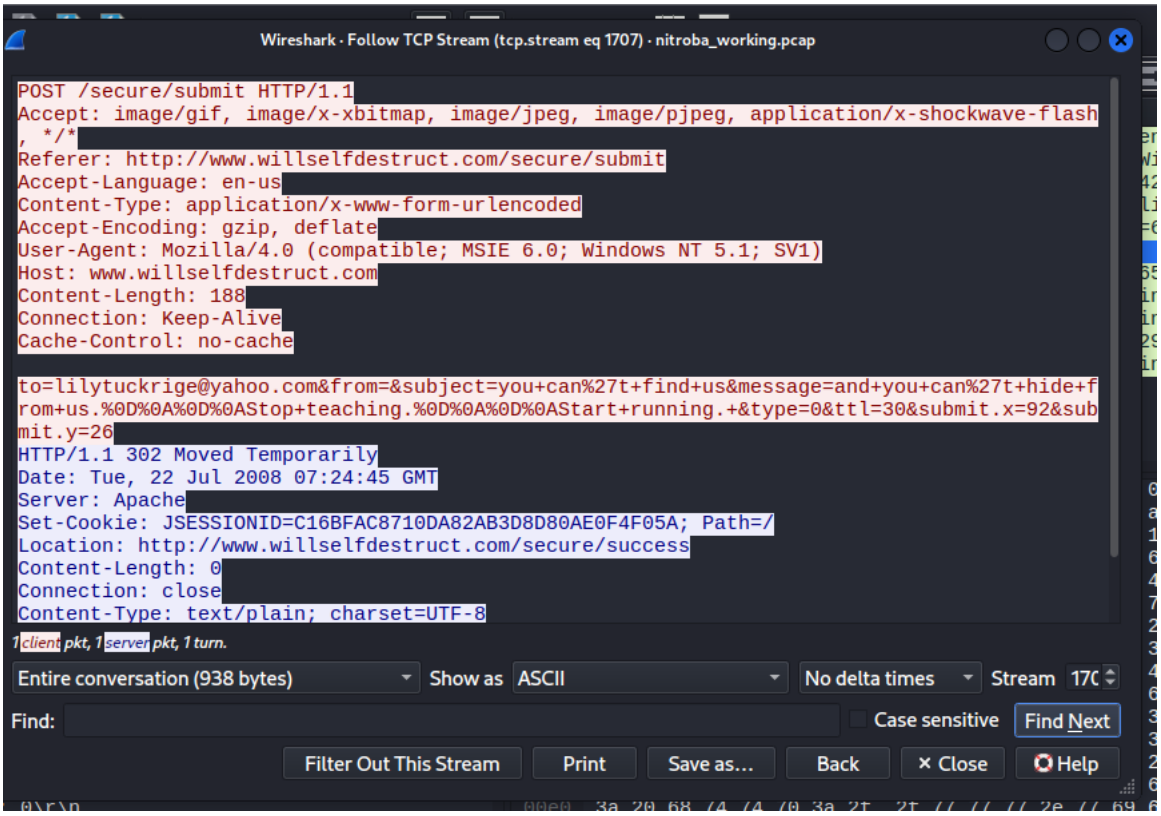
This screenshot shows the HTTP requests from **192.168.15.4** to **69.25.94.22**, including the request to /secure/submit.



Appendix B — Harassment Email TCP Stream

Screenshot: 02_harassment_email_tcp_stream.png

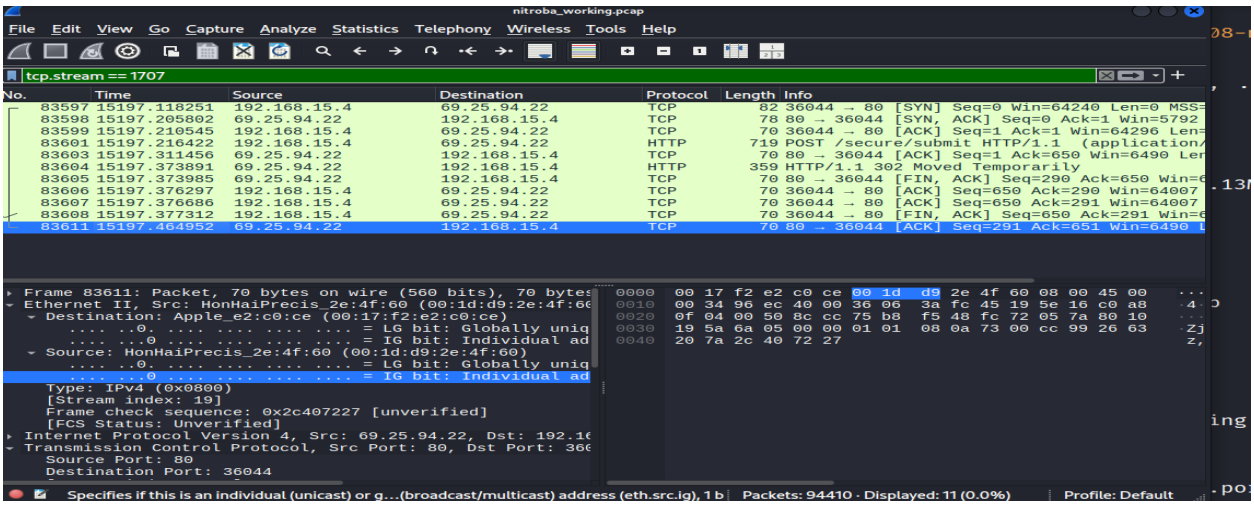
This screenshot shows the HTTP form data containing the recipient, subject and harassment message.



Appendix C — TCP Stream 1707 Transaction

Screenshot: 03_tcp_stream_1707_transaction.png

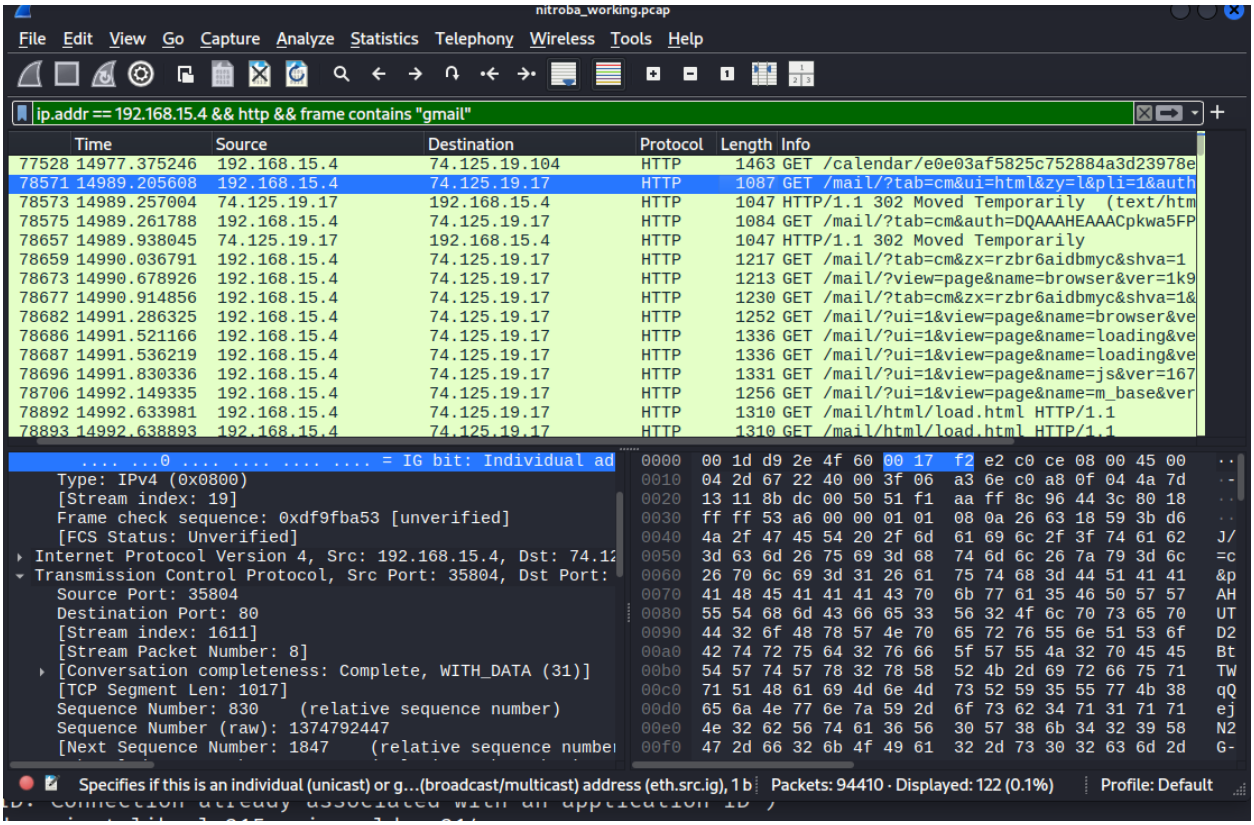
This screenshot shows the TCP connection and HTTP transaction associated with packet **83601**.



Appendix D — Gmail Account Evidence

Screenshot: 04_gmail_account_evidence.png

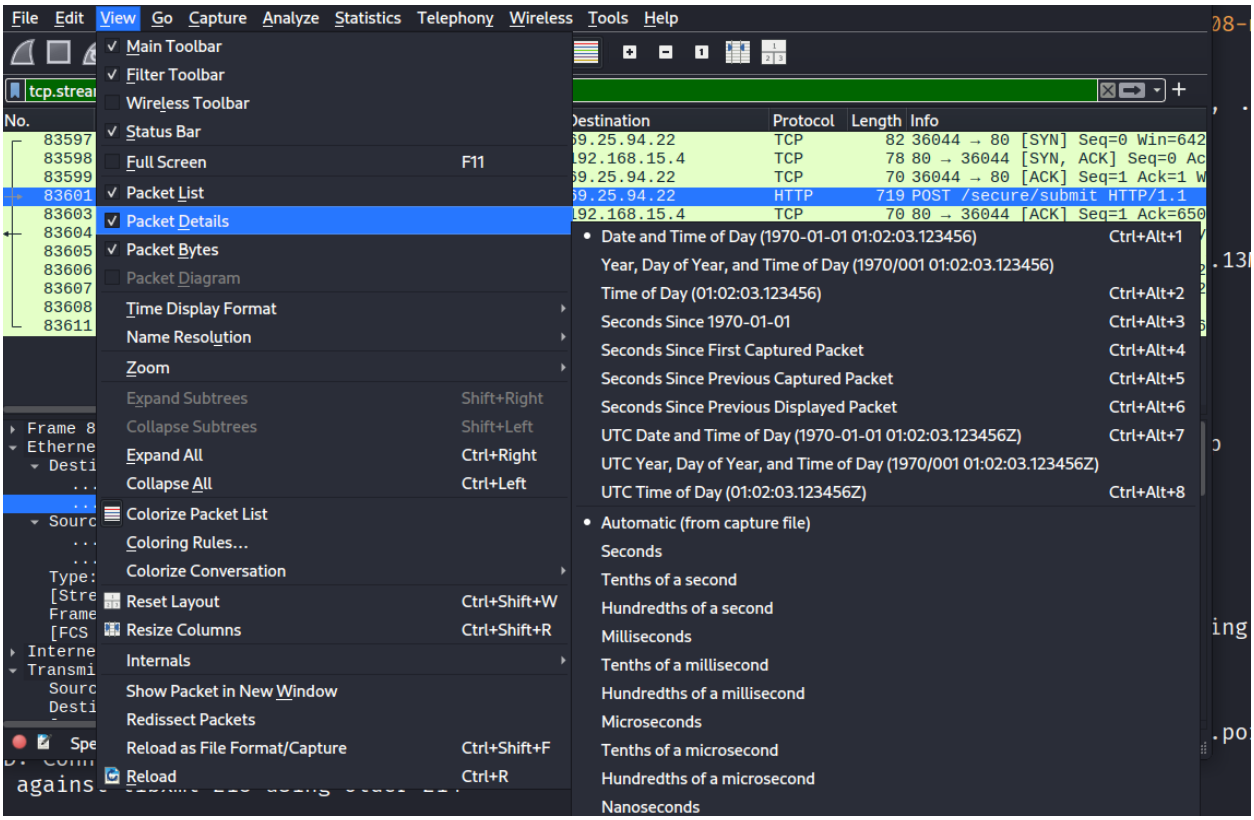
This screenshot shows packet **78571** containing the Gmail account identifier associated with the client traffic.

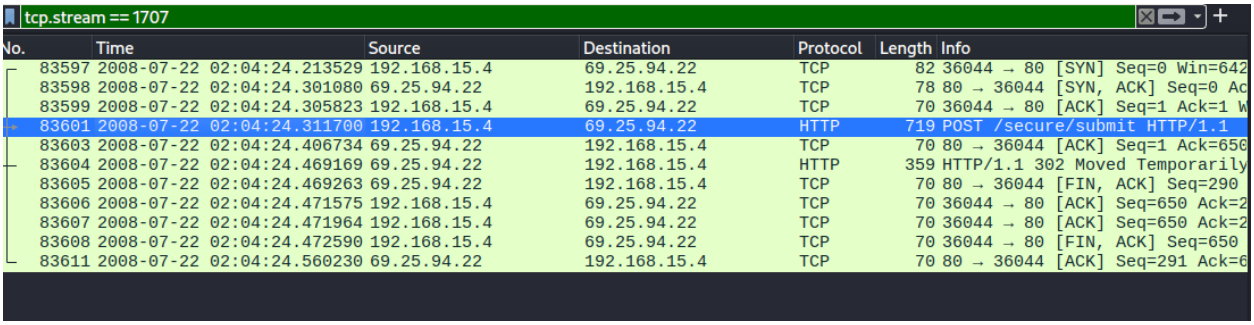


Appendix E — Harassment Timestamp

Screenshot: 05_harassment_timestamp.png

This screenshot shows the timestamp analysis for packet **83601** and supports the corrected UTC value obtained using the course-material procedure.





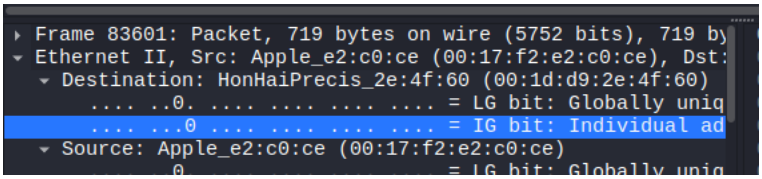
Appendix F — Device MAC Address

Screenshot: 06_device_mac_address.png

This screenshot shows the source MAC address:

00:17:f2:e2:c0:ce

Wireshark displayed the vendor label as **Apple_e2:c0:ce**.



Appendix G — Chemistry 109 Class Roster

Screenshot: 07_chemistry_109_class_roster.png

This screenshot shows the supplied **Chemistry 109 class roster**. The roster lists **Johnny Coach** as a student in the class.

This supports the association between the identified Gmail account, **jcoachj@gmail.com**, and a student in Chemistry 109.

2. Chem 109 - Class roster

Teacher: Lily Tuckrige

Students:

- Amy Smith
- Burt Greedom
- Tuck Gorge
- Ava Book
- Johnny Coach
- Jeremy Ledvkin
- Nancy Colburne
- Tamara Perkins
- Esther Pringle
- Asar Misrad
- Jenny Kant

FINAL EVIDENCE SUMMARY

The key evidence in this investigation was:

- **Client IP:** 192.168.15.4
- **Service IP:** 69.25.94.22
- **Harassment packet:** 83601
- **TCP stream:** 1707
- **Gmail identifier:** jcoachj@gmail.com
- **Identified person:** Johnny Coach
- **Chemistry class:** Chemistry 109
- **MAC address 00:17:f2:e2:c0:ce**
- **Relative packet time:** 15197.216422 seconds
- **Corrected UTC timestamp:** 1970-01-01 01:02:03.123456 UTC
- **Conclusion:** The combined evidence identified Johnny Coach as the person associated with the harassment activity. The shared/open wireless network was noted as a limitation.